JULIAN MONZAVIFAR

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EDUCATION

North Carolina State University: B.S. Mechanical Engineering — **Raleigh, NC**

- *GPA: 3.86/4.0* — *Aug 2025 - May 2029*

SKILLS & CERTIFICATIONS

- Certified SolidWorks Associate with 5+ years of CAD experience in SolidWorks and Onshape
- CNC mill/lathe, CAD, CAM, Ansys, Design for Manufacturing (DFM), Excel, MATLAB, Python

ENGINEERING & PROFESSIONAL EXPERIENCE

Weldon Steel — **Weldon, NC**

Manufacturing Engineer & Operations Intern — *May 2026 - Aug 2026*

- Designed and implemented data-collection device to monitor machine utilization while analyzing operational workflows identifying an improvement for a 30% projected increase in plasma cutter efficiency
- Collaborated with machine operators to map 80% of core fabrication workflow and created training materials to standardize manufacturing processes and improve employee onboarding
- Audited financial data to develop prediction models for revenue recognition aiding in planned capex projects

Pack Motorsports Baja SAE — **Raleigh, NC**

Drivetrain Engineer & Manufacturing Lead — *Aug 2025 - Present*

- Direct drivetrain, suspension, and brake subsystem redesigns to guarantee DFM, coordinate manufacturing timelines, and personally redesign the vehicle's U-joint axles, hubs, and uprights to reduce 20% assembly weight
- Train new members in CNC operation, CAM programming, and manual machining practices
- Programmed and machined three sets of front axle assemblies alongside the steering housing and half the CVT

Center for Additive Manufacturing and Logistics — **Raleigh, NC**

CNC & Additive Manufacturing Undergraduate Researcher — *Oct 2025- Present*

- Operate three and five axis CNC mills to machine various metals using Fusion CAM, and post machine additive manufactured components utilizing custom fixture design
- Collaborate with professors and graduate students to refine designs for manufacturability, create technical drawings, and complete machining for research projects on schedule
- Serve as technical representative, showcasing facility to faculty, prospective students, and industry sponsors

Hampton's Mobile Service — **Huntersville, NC**

Mechanic — *May 2025 - Aug 2025*

- Diagnosed mechanical and electrical issues, providing accurate service quotes and repair timelines to customers
- Performed routine maintenance and vehicle repairs to all major components in the undercarriage and engine bay

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Engineers Without Borders — **Raleigh, NC**

North Carolina Projects Member — *August 2026 - Present*

- Partner with El Futuro community nonprofit to engineer data-collection solution to quantify greenway usage, generating impact metrics required to qualify organization for state infrastructure funding
- Redesign device by integrating IR sensors for automated data-collection and designing new enclosure

Labyrinth Robotics, *FIRST* Robotics Competition Team — **Morganton, NC**

Team Mentor (Formerly Captain & Head Engineer) — *May 2025 - Present*

- Coordinate and facilitate engineering training in CAD, manufacturing, and DFM to ensure knowledge transfer
- Audit technical design viability, enforce project scope to meet deadlines, and oversee inventory and procurement
- Secured $10,000 in team funding through grant applications, outreach, and local business partnerships

HONORS & AWARDS

First Place 2026 Baja SAE Carolina Night, NCSU Dean's List, National Merit Commended Scholar, and NCSSM Outstanding Service Award (Completion of 150 hours of Community Service)